CHEM 1301 Practice Exam 3-1 — Answer Key

Worked solutions with course-material citations

Every numeric answer was recomputed by `_scripts/build_exam_3_1.py`, which builds this key and the exam from the same data. Atomic masses are periodic-table values (H 1.008, C 12.01, N 14.01, O 16.00, …), matching Ch 8 slides 5, 9, and 22. Citations name the slide (same number in the `.pdf` and `.pptx`) where the content is taught, plus the homework screenshot with the closest problem.

Score bands. 32–35 exam-ready · 27–31 review the missed topics · 26 or below rework the study guide sections cited.

Q	Ans	Answer	Type	Topic (source)
1	**d**	An increase in the total mass of the products	concept	Driving forces (Ch 7 sl. 3)
2	**a**	$PbCl_2$	concept	Solubility rules: insoluble salt (Ch 7 sl. 11)
3	**b**	$(NH_4)_3PO_4$	concept	Solubility rules: which rule wins (Ch 7 sl. 11)
4	**d**	$CaCO_3$	concept	Predicting the precipitate (Ch 7 sl. 11, 14)
5	**d**	$Ag^+(aq) + Cl^-(aq) \rightarrow AgCl(s)$	concept	Net ionic equation (Ch 7 sl. 17, 20)
6	**e**	6	calc	Balancing an acid–base neutralization (Ch 7 sl. 27)
7	**a**	KNO_3 and Na_2S	concept	Mixing with no reaction (Ch 7 sl. 11, 14)
8	**b**	$H^+(aq) + OH^-(aq) \rightarrow H_2O(l)$	concept	Strong acid + strong base net ionic (Ch 7 sl. 23, 25)
9	**d**	H_2SO_4	concept	Identifying a strong acid (Ch 7 sl. 21, 24)
10	**b**	A metal hydroxide that is very soluble in water and completely breaks up into separated ions, so it is a strong electrolyte	concept	Definition of a strong base (Ch 7 sl. 22, 25)
11	**c**	3	calc	Electrons transferred per atom (Ch 7 sl. 33)
12	**e**	oxidation–reduction and single-replacement	concept	Classifying Zn + HCl (Ch 7 sl. 40)
13	**a**	$2Fe(s) + 3Cl_2(g) \rightarrow 2FeCl_3(s)$	concept	Synthesis that is redox but not combustion (Ch 7 sl. 41–43)
14	**b**	the number of carbon atoms in exactly 12.01 g of carbon, 6.022×10^{23}	concept	Definition of the mole (Ch 8 sl. 7)
15	**c**	24.31 g of magnesium	concept	Same number of atoms (Ch 8 sl. 6)
16	**a**	0.133 mol	calc	Grams to moles (Ch 8 sl. 11)
17	**c**	7.56 g	calc	Atoms to grams (Ch 8 sl. 15)

Q	Ans	Answer	Type	Topic (source)
18	**b**	1.05×10^{23} atoms	calc	Grams to atoms (Ch 8 sl. 10, 13)
19	**e**	342.2 g/mol	calc	Molar mass with parentheses (Ch 8 sl. 22, 26)
20	**d**	98.0 g	calc	Moles to grams (Ch 8 sl. 27)
21	**a**	8.50×10^{-5} mol	calc	Milligrams to moles (Ch 8 sl. 29)
22	**c**	31.6%	calc	Mass percent (Ch 8 sl. 30–32)
23	**d**	CH_2O	concept	Empirical formula (Ch 8 sl. 39–40)
24	**e**	CCl_2F_2	calc	Percent composition to empirical formula (Ch 8 sl. 35–38)
25	**a**	$C_8H_{10}N_4O_2$	calc	Empirical to molecular formula (Ch 8 sl. 47–49)
26	**c**	Its composition by mass **and** its molar mass	concept	Data needed for a molecular formula (Ch 8 sl. 47)
27	**a**	1 mol of CO reacts with 2 mol of H_2 to produce 1 mol of CH_3OH	concept	Meaning of coefficients (Ch 9 sl. 2–3)
28	**e**	(2 mol NH_3) / (3 mol H_2)	concept	Choosing the mole ratio (Ch 9 sl. 6–7)
29	**b**	11.7 mol	calc	Mole–mole conversion (Ch 9 sl. 10)
30	**d**	the amount of product predicted from the amount of the limiting reactant	concept	Definition of theoretical yield (Ch 9 sl. 25, 28)
31	**e**	373 g	calc	Moles to grams of product (Ch 9 sl. 14, 17)
32	**e**	72.9 g	calc	Grams to grams (Ch 9 sl. 14, 19)
33	**b**	Al is the limiting reactant	calc	Identifying the limiting reactant (Ch 9 sl. 21–24)
34	**c**	160 g	calc	Theoretical yield with a limiting reactant (Ch 9 sl. 25–27)
35	**c**	77.4%	calc	Percent yield (Ch 9 sl. 30–32)

Mix: 18 concept / 17 calculation · Ch 7: 13 · Ch 8: 13 · Ch 9: 9

1. Driving forces — d) An increase in the total mass of the products

Source: Chapter 7 post, slide 3 (Driving Forces); slide 2 (Predicting Whether a Reaction Will Occur)
Homework match: HW14 — hw14ex1.png (naming the reaction type driven by forming a solid)
What it tests: Recalling the course's four driving forces: solid, water, electron transfer, gas.

Slide 3 lists exactly four driving forces: (1) formation of a solid, (2) formation of water, (3) transfer of electrons, (4) formation of a gas. Mass is conserved in every reaction, so "an increase in the total mass of the products" is not a driving force; it can't happen at all.

Why the other choices are wrong:

- a) Transfer of electrons: listed driving force #3.
- b) Formation of water: listed driving force #2.
- c) Formation of a solid: listed driving force #1.
- e) Formation of a gas: listed driving force #4.

2. Solubility rules: insoluble salt — a) $PbCl_2$

Source: Chapter 7 post, slide 11 (Solubility Rules, Table 7.1), rule 3
Homework match: HW14 — hw14ex2.png (choose the substance not likely to be soluble)
What it tests: Memorized Table 7.1, especially the named exceptions to the chloride and sulfate rules.

Rule 3: most chloride salts are soluble. **Notable exceptions are AgCl, $PbCl_2$, and Hg_2Cl_2.** $PbCl_2$ is one of the three named exceptions, so it is insoluble. The exceptions are the most likely thing to be tested here.

Why the other choices are wrong:

- b) Na_2SO_4: Rules 2 and 4: Na^+ salts and most sulfates are soluble.
- c) NH_4Cl: Rules 2 and 3: NH_4^+ salts and most chlorides are soluble.
- d) $Ba(NO_3)_2$: Rule 1: nitrates are soluble. ($BaSO_4$ is the barium exception, not the nitrate.)
- e) KNO_3: Rules 1 and 2: nitrates and K^+ salts are soluble.

3. Solubility rules: which rule wins — b) $(NH_4)_3PO_4$

Source: Chapter 7 post, slide 11 (Solubility Rules), rules 2 and 6
Homework match: HW14 — hw14ex3.png / hw14ex4.png (which rule explains solubility, e.g. ammonium sulfide)
What it tests: Applying rule 2 (Na^+, K^+, NH_4^+ salts are soluble) even when the anion is a rule-6 'insoluble' anion.

Phosphates are normally only slightly soluble (rule 6), **but** rule 2 says most salts of Na^+, K^+, and NH_4^+ are soluble. The cation rule wins, so $(NH_4)_3PO_4$ is soluble. HW14 does the same thing with ammonium sulfide and ammonium carbonate, where the answer is "most salts of NH_4^+ are soluble."

Why the other choices are wrong:

- a) Ag_2S: Rule 6: sulfides are only slightly soluble.
- c) $Mg(OH)_2$: Rule 5: most hydroxides are only slightly soluble (only NaOH and KOH are listed as soluble).
- d) $CaCO_3$: Rule 6: carbonates are only slightly soluble, and Ca^{2+} isn't a rule-2 cation.
- e) $BaSO_4$: Rule 4 exception: $BaSO_4$ is insoluble.

4. Predicting the precipitate — d) $CaCO_3$

Source: Chapter 7 post, slide 14 (How to Predict Precipitates), slide 11 (rules 1, 2, 6), slide 10 (a solid must contain cations and anions)
Homework match: HW15 — hw15ex4.png (molecular equations for precipitation, e.g. $K_2CO_3 + MgCl_2 \rightarrow MgCO_3(s)$)
What it tests: The three-step method: write the ions, swap the anions, check each possible product against the solubility rules.

Step 1: the ions present are Ca^{2+}, NO_3^-, Na^+, CO_3^{2-}. Step 2: swapping anions gives the possible solids $CaCO_3$ and $NaNO_3$. Step 3: $NaNO_3$ is soluble (rules 1, 2), and $CaCO_3$ is only slightly soluble (rule 6), so the solid is **$CaCO_3$**.

$Ca(NO_3)_2(aq) + Na_2CO_3(aq) \rightarrow CaCO_3(s) + 2NaNO_3(aq)$

Why the other choices are wrong:

- a) $CaNa_2$: Two cations can't combine. A solid must contain both a cation and an anion (slide 10).
- b) No precipitate forms: $CaCO_3$ is only slightly soluble (rule 6), so a solid does form.
- c) $NaNO_3$: Rules 1 and 2: $NaNO_3$ is soluble and stays as Na^+ and $NO_3{}^-$ ions.
- e) $Na(NO_3)_2$: Not a valid neutral formula, and it isn't what you get from swapping anions.

5. Net ionic equation — d) $Ag^+(aq) + Cl^-(aq) \rightarrow AgCl(s)$

Source: Chapter 7 post, slide 20 (Practice: NaCl + $AgNO_3$), slides 15–18 (molecular / complete ionic / net ionic)
Homework match: HW15 — hw15ex1.png (definition of net ionic equation) and hw15ex2.png (Sr^{2+} + $CrO_4{}^{2-}$ net ionic)
What it tests: Telling the three equation types apart; a net ionic equation leaves out the spectator ions.

Molecular: $NaCl(aq) + AgNO_3(aq) \rightarrow AgCl(s) + NaNO_3(aq)$. Complete ionic: write every strong electrolyte as ions. Na^+ and $NO_3{}^-$ appear unchanged on both sides, so they are spectator ions. Cancel them to get the net ionic equation: $\mathbf{Ag^+(aq) + Cl^-(aq) \rightarrow AgCl(s)}$ (slide 17: "only those components that are directly involved").

Why the other choices are wrong:

- a) $Ag^+(aq) + Cl^-(aq) + Na^+(aq) + NO_3{}^-(aq) \rightarrow AgCl(s) + Na^+(aq) + NO_3{}^-(aq)$: This is the complete ionic equation. It still has the spectator ions.
- b) $Na^+(aq) + NO_3{}^-(aq) \rightarrow NaNO_3(s)$: $NaNO_3$ is soluble (rules 1, 2), so these are spectator ions.
- c) $NaCl(aq) + AgNO_3(aq) \rightarrow AgCl(s) + NaNO_3(aq)$: This is the molecular equation (slide 15).
- e) $Ag^+(aq) + Cl^-(aq) \rightarrow AgCl(aq)$: AgCl is an exception to rule 3. It precipitates, so its state is (s).

6. Balancing an acid–base neutralization — e) 6

Source: Chapter 7 post, slide 27 (Practice: $H_3PO_4 + Ca(OH)_2 \rightarrow Ca_3(PO_4)_2 + H_2O$)
Homework match: HW22 — balancing equations with lowest whole-number coefficients ($KrF_2 + H_2O$; $Al + H_2SO_4$)
What it tests: Balancing an acid–base reaction: polyatomic ions go first as units, then H and O.

Balance Ca first: 3 $Ca(OH)_2$. Balance PO_4 as a unit: 2 H_3PO_4. Now count H and O outside PO_4: 6 H from the acid + 6 OH from the base make **6 $\mathbf{H_2O}$**.

$$2\,H_3PO_4 + 3\,Ca(OH)_2 \rightarrow Ca_3(PO_4)_2 + 6\,H_2O$$

Check: H is 6 + 6 = 12 on the left and 12 on the right. O is 8 + 6 = 14 on the left and 8 + 6 = 14 on the right.

Why the other choices are wrong:

- a) 1: Leaves the equation unbalanced.
- b) 2: This is the coefficient of H_3PO_4.
- c) 3: This is the coefficient of $Ca(OH)_2$.
- d) 4: Doesn't balance H or O.

7. Mixing with no reaction — a) KNO_3 and Na_2S

Source: Chapter 7 post, slide 14 (How to Predict Precipitates), slide 11 (Solubility Rules)
Homework match: HW15 — hw15ex3.png (does a reaction occur: potassium nitrate + sodium sulfide → NR; aluminum chloride + potassium hydroxide)
What it tests: Swapping anions and checking both possible products. If both are soluble, there is no reaction (NR).

KNO_3 + Na_2S: the possible products from swapping anions are K_2S and $NaNO_3$. K_2S is soluble because K^+ salts are soluble (rule 2), and $NaNO_3$ is soluble (rules 1, 2). Nothing precipitates, so the answer is **NR**. Every other pair makes one insoluble product.

Why the other choices are wrong:

- b) $AlCl_3$ and KOH: Forms $Al(OH)_3$(s) (rule 5).
- c) $MgCl_2$ and K_2CO_3: Forms $MgCO_3$(s) (rule 6).
- d) $BaCl_2$ and Na_2SO_4: Forms $BaSO_4$(s) (rule 4 exception).
- e) $AgNO_3$ and NaCl: Forms AgCl(s) (rule 3 exception).

8. Strong acid + strong base net ionic — b) $H^+(aq) + OH^-(aq) \rightarrow H_2O(l)$

Source: Chapter 7 post, slide 23 (Acid–Base Reactions), slide 25 (principle 4), slide 26 (principles 5–6)
Homework match: HW16 — hw16ex2.png (the same net ionic process for any strong acid/strong base)
What it tests: The net ionic equation for every strong acid + strong base reaction is the formation of water.

Slide 25, principle 4: "The net ionic equation for the reaction of a strong acid and a strong base … is always the same: it shows the production of water." The cation of the base and the anion of the acid are spectators, so the equation is $\mathbf{H^+(aq) + OH^-(aq) \rightarrow H_2O(l)}$.

Why the other choices are wrong:

- a) $HCl(aq) + NaOH(aq) \rightarrow NaCl(aq) + H_2O(l)$: A molecular equation for one specific acid and base.
- c) $H^+(aq) + Cl^-(aq) \rightarrow HCl(aq)$: Strong acids completely dissociate. They don't re-form.
- d) $2H^+(aq) + O^{2-}(aq) \rightarrow H_2O(l)$: Strong bases release OH^-, not O^{2-}.
- e) $Na^+(aq) + Cl^-(aq) \rightarrow NaCl(s)$: The salt stays dissolved (slide 26, principle 5), so Na^+ and Cl^- are spectators.

9. Identifying a strong acid — d) H_2SO_4

Source: Chapter 7 post, slide 24 (Summary of Strong Acids and Strong Bases, principle 1), slide 21 (Arrhenius Acids)
Homework match: HW16 — hw16ex2.png (salts formed by strong acid/strong base reactions: HI, HNO_3, HCl)
What it tests: Memorizing the course's list of strong acids (HCl, HNO_3, H_2SO_4) and the definition of a strong acid.

Slide 24, principle 1: "The common strong acids are aqueous solutions of **HCl, HNO_3, and H_2SO_4**." A strong acid completely dissociates (ionizes) in water (principle 2). Watch HNO_2 (nitrous) versus HNO_3 (nitric). Only nitric is strong.

Why the other choices are wrong:

- a) H_3PO_4: Phosphoric acid (slide 27) is not on the strong-acid list.
- b) HF: Not on the strong-acid list. HCl is the strong hydrogen halide.
- c) HNO_2: Nitrous acid (slide 28) is not on the strong-acid list. Don't mix it up with HNO_3.
- e) $HC_2H_3O_2$: Acetic acid is not on the strong-acid list.

10. Definition of a strong base — b) A metal hydroxide that is very soluble in water and completely breaks up into separated ions, so it is a strong electrolyte

Source: Chapter 7 post, slide 25 (principle 3), slide 22 (Arrhenius Bases), slide 7 (Strong Electrolytes)
Homework match: HW16 — hw16ex1.png (what is a strong base; are strong bases strong electrolytes?)
What it tests: Course definitions: Arrhenius base, strong base, strong electrolyte.

Slide 25, principle 3: "A strong base is a metal hydroxide compound that is very soluble in water. The most common strong bases are NaOH and KOH, which completely break up into separated ions." Because every unit gives separated ions, strong bases are **strong electrolytes** (slide 7 definition).

Why the other choices are wrong:

- a) Any compound containing OH, whether or not it dissolves: Most hydroxides are only slightly soluble (rule 5). Those aren't strong bases.
- c) A metal hydroxide that produces OH^- ions but is not a strong electrolyte: Strong bases are strong electrolytes (HW16).
- d) A substance that accepts electrons from a metal: That describes the nonmetal in a redox reaction (slide 31).
- e) A substance that produces H^+ ions when dissolved in water: That is an Arrhenius acid (slide 21).

11. Electrons transferred per atom — c) 3

Source: Chapter 7 post, slide 33 (Oxidation–Reduction Reactions Practice: $2Al + N_2 \rightarrow 2AlN$), slides 30–31
Homework match: HW16 — hw16ex3.png ($Ba + Cl_2 \rightarrow BaCl_2$: Ba loses two electrons)
What it tests: Metal + nonmetal is always redox. Get the ion charges from the ionic product.

Metal + nonmetal makes an ionic compound, so this is redox (slide 31). In AlN, nitrogen forms N^{3-} (group 15 gains 3), so aluminum must be Al^{3+} for the compound to be neutral. Each Al atom **loses 3 electrons**, and each N atom gains 3. Total electrons moved per equation $= 2 \times 3 = 6$.

Why the other choices are wrong:

- a) 1: That would give Al^+, but AlN needs Al^{3+} to balance N^{3-}.
- b) 2: The subscript on N_2 has nothing to do with the charge.
- d) 4: No combination of the ion charges gives 4.
- e) 6: That is the total for 2 Al atoms, not per atom.

12. Classifying Zn + HCl — e) oxidation–reduction and single-replacement

Source: Chapter 7 post, slide 40 (Single-Replacement Reactions), slide 45 (Summary of Classes of Reactions)
Homework match: HW17 — hw17ex1.png (classify HCl(aq) + Fe(s) → H_2(g) + $FeCl_2$(aq) as oxidation–reduction)
What it tests: A reaction can be in more than one class. A single-replacement reaction is always redox.

Zn goes from a neutral atom to Zn^{2+} (loses 2 e^-), and H^+ becomes H_2 (gains electrons), so this is **redox**. One anion (Cl^-) is passed from H^+ to Zn^{2+}, which is the slide 40 definition of a **single-replacement** reaction.

Why the other choices are wrong:

- a) combustion and synthesis: No O_2 is involved, and it doesn't make one compound from simpler ones.
- b) decomposition: Two reactants combine into new products. Nothing is broken down.
- c) precipitation and double-displacement: No solid forms, and only one anion (Cl^-) changes partners.
- d) acid–base only: No OH^- is involved, and H^+ ends up in H_2, not in water.

13. Synthesis that is redox but not combustion — a) 2Fe(s) + 3Cl_2(g) → 2$FeCl_3$(s)

Source: Chapter 7 post, slide 42 (Synthesis), slide 41 (Combustion), slide 43 (Decomposition), slide 31 (metal + nonmetal = redox)
Homework match: HW17 — hw17ex3.png (synthesis also combustion vs. synthesis also redox but not combustion)
What it tests: Telling synthesis, combustion, and decomposition apart, and seeing that one reaction can belong to several classes.

Synthesis makes one compound from simpler materials (slide 42). Fe + Cl_2 → $FeCl_3$ qualifies, and it is metal + nonmetal, so it is redox (slide 31). No O_2 is involved, so it is not combustion (slide 41). C + O_2 is also a synthesis, but it burns in oxygen, so it counts as combustion.

Why the other choices are wrong:

- b) 2HgO(s) → 2Hg(l) + O_2(g): Decomposition (slide 43).
- c) CH_4(g) + 2O_2(g) → CO_2(g) + 2H_2O(g): Combustion, not synthesis. It makes two products.
- d) NaOH(aq) + HCl(aq) → NaCl(aq) + H_2O(l): Acid–base, with no electron transfer.
- e) C(s) + O_2(g) → CO_2(g): Synthesis and redox, but it burns in O_2, so it is also combustion.

14. Definition of the mole — b) the number of carbon atoms in exactly 12.01 g of carbon, 6.022 × 10^{23}

Source: Chapter 8 post, slide 7 (The Mole); slide 5 (Atomic Mass Units)
Homework match: HW18 — the atoms → grams conversion factor (1 mol / 6.022 × 10^{23} atoms)
What it tests: The course's wording of the mole and Avogadro's number, and how they differ from the amu.

Slide 7: "The mole (mol) can be defined as the number equal to the number of carbon atoms in 12.01 grams of carbon … 6.022×10^{23}. This number is called Avogadro's number."

Why the other choices are wrong:

- a) the mass of one atom of carbon, 12.01 g: 12.01 g is the mass of a *mole* of C. One atom is 12.01 amu.
- c) the number of atoms in 1.000 g of any element: Different elements have different atomic masses, so 1 g contains different numbers of atoms.
- d) the number of molecules in 1 L of any substance: That isn't how the mole is defined.
- e) 1.66×10^{-24} g: That is 1 amu (slide 5).

15. Same number of atoms — c) 24.31 g of magnesium

Source: Chapter 8 post, slide 6 (Elements: 26.98 g Al vs 63.55 g Cu), slide 9 (Table 8.2, 1-mole samples)
Homework match: HW18 — comparing samples (Co atoms vs Mn moles: which contains more atoms?)
What it tests: Samples whose masses are in the same ratio as their atomic masses contain the same number of atoms.

26.98 g Al = 1 mol Al = 6.022×10^{23} atoms. The same number of Mg atoms is 1 mol Mg = **24.31 g**. Slide 6: "samples in which the ratio of the masses is the same as the ratio of the masses of the individual atoms always contain the same number of atoms."

Why the other choices are wrong:

- a) 13.49 g of copper: This is half the Al mass. Equal atoms needs 63.55 g Cu (slide 6).
- b) 26.98 g of magnesium: Equal masses of different elements do NOT have equal numbers of atoms.
- d) 6.022 g of hydrogen: 1 mol of H is 1.008 g, so 6.022 g H is about 6 mol.
- e) 12.01 g of magnesium: 12.01 g is 1 mol of *carbon*, not magnesium.

16. Grams to moles — a) 0.133 mol

Source: Chapter 8 post, slide 11 (Mole Conversion Practice: 3.60 g Al)
Homework match: HW18 — hw18ex1.png (moles from mass: neon, silver, uranium)
What it tests: Grams → moles: divide by the molar mass from the periodic table.

$$3.60 \text{ g Al} \times \frac{1 \text{ mol Al}}{26.98 \text{ g Al}} = 0.133 \text{ mol Al}$$

Why the other choices are wrong:

- b) 1.33 mol: Decimal slip (off by 10).
- c) 7.49 mol: Inverted: 26.98 / 3.60.
- d) 97.1 mol: Multiplied by the molar mass instead of dividing.
- e) 8.04×10^{22} mol: That is the number of atoms, not moles.

17. Atoms to grams — c) 7.56 g

Source: Chapter 8 post, slide 15 (Convert atoms to grams: 2.31×10^{22} atoms of gold), slide 10 (Mole Conversion Example)
Homework match: HW18 — atoms → grams setup ((1 mol / 6.022×10^{23} atoms) × (g / 1 mol))
What it tests: Atoms → moles (÷ Avogadro's number) → grams (× molar mass).

$$2.31 \times 10^{22} \text{ atoms} \times \frac{1 \text{ mol}}{6.022 \times 10^{23} \text{ atoms}} \times \frac{196.97 \text{ g}}{1 \text{ mol}} = 0.03836 \text{ mol} \times 196.97 = 7.56 \text{ g Au}$$

Why the other choices are wrong:

- a) 1.95×10^{-4} g: Divided by the molar mass instead of multiplying.
- b) 0.0384 g: Stopped at moles (0.0384 mol).
- d) 5.13×10^{3} g: Inverted the mole value: 196.97 / 0.0384.
- e) 4.55×10^{24} g: Multiplied atoms by molar mass without dividing by Avogadro's number.

18. Grams to atoms — b) 1.05×10^{23} atoms

Source: Chapter 8 post, slide 10 (Mole Conversion Example: 0.500 g H → atoms), slides 13–14 (Atom Conversion Practice)
Homework match: HW18 — hw18ex3.png (number of atoms in 3.00 g Ag, 6.50 kg Mg, …)
What it tests: Grams → moles → atoms, following slide 10's two-step pattern.

$$4.25 \text{ g Mg} \times \frac{1 \text{ mol}}{24.31 \text{ g}} = 0.1748 \text{ mol}; \quad 0.1748 \text{ mol} \times 6.022 \times 10^{23} \ \tfrac{\text{atoms}}{\text{mol}} = 1.05 \times 10^{23} \text{ atoms}$$

Why the other choices are wrong:

- a) 1.05×10^{22} atoms: Exponent slip: 10^{23} is right, not 10^{22}.
- c) 2.56×10^{24} atoms: Skipped the grams → moles step (treated 4.25 g as 4.25 mol).
- d) 3.44×10^{24} atoms: Inverted the mole step: 24.31 / 4.25.
- e) 6.22×10^{25} atoms: Multiplied by the molar mass instead of dividing.

19. Molar mass with parentheses — e) 342.2 g/mol

Source: Chapter 8 post, slide 22 (Molar Mass: sum of component atoms), slide 26 (LA problem: $Ba(OH)_2$ with parentheses)
Homework match: HW19 — hw19ex1.png (write formula and molar mass: iron(II) chloride, copper(II) sulfate, rubidium chloride)
What it tests: Molar mass = sum of atomic masses × subscripts. A subscript outside parentheses multiplies everything inside.

Atom count: 2 Al, 3 S, 12 O.

$$2(26.98) + 3(32.07) + 12(16.00) = 53.96 + 96.21 + 192.00 = 342.17 \approx 342.2 \text{ g/mol}$$

Why the other choices are wrong:

- a) 123.0 g/mol: Ignored every subscript outside the parentheses ($AlSO_4$).
- b) 150.0 g/mol: Dropped the 3 outside the parentheses (Al_2SO_4).
- c) 214.2 g/mol: Applied the 3 to S only, not to O.
- d) 315.2 g/mol: Dropped the 2 on Al.

20. Moles to grams — d) 98.0 g

Source: Chapter 8 post, slide 27 (Calculating Mass from Moles Practice: NaOH)
Homework match: HW18 — hw18ex2.png (mass in grams from moles: Li, Al, Pb)
What it tests: Moles → grams: multiply by the molar mass.

Molar mass NaOH = 22.99 + 16.00 + 1.008 = 40.00 g/mol.

$$2.45 \text{ mol} \times 40.00 \ \tfrac{\text{g}}{\text{mol}} = 98.0 \text{ g}$$

Why the other choices are wrong:

- a) 0.0613 g: Divided by the molar mass instead of multiplying.
- b) 16.3 g: Divided the molar mass by the moles.
- c) 56.3 g: Used only the mass of Na (22.99) as the molar mass.
- e) 143 g: Used the molar mass of NaCl (58.44).

21. Milligrams to moles — a) 8.50×10^{-5} mol

Source: Chapter 8 post, slide 29 (LA problem 1: 9.43 mg of calcium chloride), slide 28 (moles from mass)
Homework match: HW19 — hw19ex2.png (moles in 21.9 mg of nitrogen dioxide, …)
What it tests: Unit conversion (mg → g) before the mole conversion, plus writing the correct ionic formula.

$CaCl_2$: 40.08 + 2(35.45) = 110.98 g/mol. Convert mg to g first: 9.43 mg = 9.43×10^{-3} g.

$$\frac{9.43 \times 10^{-3} \text{ g}}{110.98 \text{ g/mol}} = 8.50 \times 10^{-5} \text{ mol}$$

Why the other choices are wrong:

- b) 1.25×10^{-4} mol: Wrong formula: CaCl (forgot Ca^{2+} needs two Cl^-).
- c) 8.50×10^{-2} mol: Didn't convert mg to g (treated 9.43 mg as 9.43 g).
- d) 1.05 mol: Multiplied by the molar mass.
- e) 5.12×10^{19} mol: That is the number of formula units, not moles.

22. Mass percent — c) 31.6%

Source: Chapter 8 post, slide 30 (Mass Percent definition), slide 31 (worked ethanol example), slide 32 (Practice: H_3PO_4)
Homework match: HW20 — hw20ex1.png / hw20ex3.png (percent by mass of each element: $HClO_2$, PCl_5)
What it tests: Mass percent = (mass of element in 1 mol) / (molar mass) × 100%.

Molar mass $H_3PO_4 = 3(1.008) + 30.97 + 4(16.00) = 97.99$ g/mol.

$$\%\text{P} = \frac{30.97}{97.99} \times 100\% = 31.6\%$$

(For a full check: H = 3.09%, O = 65.31%, and 3.09 + 31.6 + 65.31 = 100%.)

Why the other choices are wrong:

- a) 3.09%: This is the mass percent of H.
- b) 12.5%: 1 P out of 8 atoms is an atom fraction, not a mass fraction.
- d) 48.4%: Divided by the mass of O only, not the whole molar mass.
- e) 65.3%: This is the mass percent of O.

23. Empirical formula — d) CH_2O

Source: Chapter 8 post, slide 39 (Empirical and Molecular Formulas: $C_6H_{12}O_6 \rightarrow CH_2O$), slide 48
Homework match: HW20 — empirical formula for benzoic acid, $C_7H_6O_2$ (already in simplest form)
What it tests: Empirical formula = smallest whole-number ratio; molecular formula = $(\text{empirical})_n$.

Divide every subscript by their greatest common factor, 6: $C_{6/6}H_{12/6}O_{6/6} = \mathbf{CH_2O}$. Slide 39 uses glucose as the example: "The empirical formula for both is CH_2O."

Why the other choices are wrong:

- a) $C_3H_6O_3$: Same ratio but not reduced to the smallest whole numbers.
- b) CHO: The H:C ratio is 2:1, not 1:1.
- c) $C_2H_4O_2$: Same ratio but not the smallest whole numbers (÷2 only; ÷6 is possible).
- e) $C_6H_{12}O_6$: That is the molecular formula.

24. Percent composition to empirical formula — e) CCl_2F_2

Source: Chapter 8 post, slide 38 (Percent composition to formula: C 9.94%, Cl 58.6%, F 31.4%), slides 35–37 (worked steps), slides 42–43
Homework match: HW21 — empirical formula from percent composition (Mg/S/O → $MgSO_4$; Si/Cl/Br → $SiCl_2Br_2$)
What it tests: The slide 35–37 procedure: % → g (100 g sample) → mol → divide by the smallest.

Assume 100 g, so the percentages become grams.

	grams	÷ molar mass	moles	÷ smallest (0.828)
C	9.94	12.01	0.828	1.00

	grams	÷ molar mass	moles	÷ smallest (0.828)
Cl	58.6	35.45	1.653	2.00
F	31.4	19.00	1.653	2.00

Empirical formula $\mathbf{CCl_2F_2}$.

Why the other choices are wrong:

- a) CClF: Didn't divide by the smallest number of moles; the ratio is 1:2:2.
- b) $CClF_2$: Cl moles: 58.6 / 35.45 = 1.65, which is 2× the C moles, not 1×.
- c) CCl_2F: F moles (1.65) equal Cl moles, so F also gets a subscript of 2.
- d) $C_2Cl_3F_3$: Doesn't match the mole ratio 1 : 2 : 2.

25. Empirical to molecular formula — a) $C_8H_{10}N_4O_2$

Source: Chapter 8 post, slide 49 (LA Practice: $C_4H_5N_2O$, MM 194.14), slide 48 ($(\text{empirical formula})_n$ = molecular formula), slide 47
Homework match: HW21 — molecular formula from empirical formula + molar mass (CH → C_2H_2; C_3H_7NO → $C_6H_{14}N_2O_2$)
What it tests: n = molar mass / empirical formula mass; multiply every subscript by n.

Empirical formula mass: 4(12.01) + 5(1.008) + 2(14.01) + 16.00 = 97.10 g/mol.

$$n = \frac{194.14}{97.10} = 2.00 \quad \Rightarrow \quad (C_4H_5N_2O)_2 = C_8H_{10}N_4O_2$$

(This is caffeine.)

Why the other choices are wrong:

- b) $C_{12}H_{15}N_6O_3$: n = 3 would give 291 g/mol.
- c) $C_{16}H_{20}N_8O_4$: n = 4 would give 388 g/mol.
- d) $C_4H_5N_2O$: Assumes n = 1. The empirical mass is 97.10, not 194.14.
- e) $C_8H_{10}N_4O$: Every subscript, including O, has to be multiplied by n.

26. Data needed for a molecular formula — c) Its composition by mass and** its molar mass**

Source: Chapter 8 post, slide 47 (From Empirical to Molecular Formula: 'To obtain the molecular formula we must know the molar mass')
Homework match: HW21 — hw21ex1.png (what evidence is needed for empirical vs molecular formula)
What it tests: Empirical formula needs composition; molecular formula needs composition plus molar mass.

Slide 47: from the masses or mass percentages "we can calculate the empirical formula but not the molecular formula. To obtain the molecular formula we must know the molar mass." So you need both, which is exactly the HW21 select-all answer.

Why the other choices are wrong:

- a) Its composition by mass only: That is enough for the empirical formula only (slide 47).
- b) The mass of the sample only: Gives no information about which elements are present or in what ratio.
- d) The mass of the sample and its melting point: Neither fixes the atom ratio.
- e) Its molar mass only: Many formulas share a molar mass. You need the composition too.

27. Meaning of coefficients — a) 1 mol of CO reacts with 2 mol of H_2 to produce 1 mol of CH_3OH

Source: Chapter 9 post, slide 3 (Table 9.1, Methanol Production), slide 2 (Information Given by Chemical Equations)
Homework match: HW22 — interpreting $H_2 + F_2 \rightarrow 2HF$ in molecules and in moles
What it tests: Coefficients give relative numbers of molecules or moles, never grams.

Slide 2: "The coefficients in a balanced equation give us the relative numbers of molecules." Table 9.1 reads the same equation as 1 molecule CO + 2 molecules H_2 → 1 molecule CH_3OH, and equally as **1 mol CO + 2 mol H_2 → 1 mol CH_3OH**.

Why the other choices are wrong:

- b) 1 molecule of CO reacts with 2 atoms of H to produce 1 molecule of CH_3OH: The 2 counts H_2 *molecules* (4 H atoms).
- c) 3 g of reactants produce 1 g of product: Mass is conserved. Coefficients don't describe grams.
- d) 1 g of CO reacts with 2 g of H_2 to produce 1 g of CH_3OH: Coefficients are mole (particle) ratios, not mass ratios.
- e) 1 mol of CO reacts with 1 mol of H_2 to produce 1 mol of CH_3OH: Ignores the coefficient 2 on H_2.

28. Choosing the mole ratio — e) (2 mol NH_3) / (3 mol H_2)

Source: Chapter 9 post, slide 7 (Mole Ratios Practice: $N_2 + 3H_2 \rightarrow 2NH_3$), slide 6 (Mole Ratios)
Homework match: HW22 — choosing the mole ratio for $4B + 3O_2 \rightarrow 2B_2O_3$
What it tests: Writing mole ratios from coefficients and orienting them so the units cancel.

From the coefficients, 3 mol H_2 ⇔ 2 mol NH_3. To convert *from* H_2 *to* NH_3, put H_2 on the bottom so it cancels: mol H_2 × (2 mol NH_3 / 3 mol H_2).

Why the other choices are wrong:

- a) (2 mol NH_3) / (1 mol N_2): This ratio starts from N_2, not H_2.
- b) (3 mol H_2) / (2 mol NH_3): Upside down. The unit you want (NH_3) goes on top.
- c) (17.03 g NH_3) / (2.016 g H_2): That is a ratio of molar masses, not a mole ratio.
- d) (1 mol NH_3) / (3 mol H_2): The coefficient of NH_3 is 2.

29. Mole–mole conversion — b) 11.7 mol

Source: Chapter 9 post, slide 10 (Mole–Mole Conversions Practice: $Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$, 17.5 mol CO)
Homework match: HW22 — moles of product from moles (e.g. Fe_2O_3 + Al: mol Fe from 0.660 mol Al_2O_3)
What it tests: mol given × (coefficient wanted / coefficient given).

$$17.5 \text{ mol CO} \times \frac{2 \text{ mol Fe}}{3 \text{ mol CO}} = 11.7 \text{ mol Fe}$$

Why the other choices are wrong:

- a) 5.83 mol: Used 1/3 (forgot the 2 on Fe).
- c) 17.5 mol: Assumed a 1:1 ratio.
- d) 26.3 mol: Inverted the ratio (3/2).
- e) 35.0 mol: Multiplied by 2 only (forgot the 3 on CO).

30. Definition of theoretical yield — d) the amount of product predicted from the amount of the limiting reactant

Source: Chapter 9 post, slide 25 (Theoretical Yield), slide 28 (Actual Yield), slide 30 (Percent Yield)
Homework match: HW24 — theoretical yield vs percent yield (XeF_4, $Na_2S_2O_3 \cdot 5H_2O$)
What it tests: Telling theoretical, actual, and percent yield apart (course definitions).

Slide 25: the amount of product calculated from the limiting reactant "is called the theoretical yield … the amount of product predicted from the amounts of reactants used." Slide 28 calls it "the maximum amount of product that can be produced."

Why the other choices are wrong:

- a) the total mass of all reactants added: Reactants left in excess don't become product.
- b) the amount of product actually obtained in the laboratory: That is the actual yield (slide 28).
- c) (actual yield ÷ theoretical yield) × 100%: That is the percent yield (slide 30).
- e) the amount of product predicted from the amount of the excess reactant: The excess reactant doesn't run out, so it doesn't limit the product.

31. Moles to grams of product — e) 373 g

Source: Chapter 9 post, slide 17 (Stoichiometry Practice: $2SO_2 + O_2 \rightarrow 2SO_3$, 2.33 mol O_2), slide 14 (Steps for calculating masses)
Homework match: HW23 — mass of product from mass of reactant ($CaC_2 \rightarrow C_2H_2$; $Fe + Cl_2 \rightarrow FeCl_3$)
What it tests: mol given → mol wanted (mole ratio) → grams (molar mass).

$$2.33 \text{ mol } O_2 \times \frac{2 \text{ mol } SO_3}{1 \text{ mol } O_2} \times \frac{80.07 \text{ g } SO_3}{1 \text{ mol } SO_3} = 4.66 \times 80.07 = 373 \text{ g}$$

Why the other choices are wrong:

- a) 4.66 g: That is the moles of SO_3; it still has to be converted to grams.
- b) 149 g: 1:1 ratio and the molar mass of SO_2.
- c) 187 g: Assumed 1:1 (forgot the 2 mol SO_3 per mol O_2).
- d) 299 g: Used the molar mass of SO_2 instead of SO_3.

32. Grams to grams — e) 72.9 g

Source: Chapter 9 post, slide 19 (Stoichiometry Practice 3: $4NH_3 + 5O_2 \rightarrow 4NO + 6H_2O$, 54.7 g NO), slide 14
Homework match: HW23 — grams of bromine from 34.0 g of bromine trifluoride
What it tests: The full g → mol → mol → g path from slide 14.

$$54.7 \text{ g NO} \times \frac{1 \text{ mol}}{30.01 \text{ g}} = 1.823 \text{ mol NO}$$

$$1.823 \text{ mol NO} \times \frac{5 \text{ mol } O_2}{4 \text{ mol NO}} = 2.278 \text{ mol } O_2; \quad 2.278 \times 32.00 = 72.9 \text{ g } O_2$$

Why the other choices are wrong:

- a) 2.28 g: That is the moles of O_2, not grams.
- b) 46.7 g: Inverted the mole ratio (4/5).
- c) 54.7 g: Assumed the grams of O_2 equal the grams of NO.
- d) 58.3 g: Assumed a 1:1 mole ratio.

33. Identifying the limiting reactant — b) Al is the limiting reactant

Source: Chapter 9 post, slide 24 (Limiting Reactants Practice 2: $4Al + 3O_2$, 17.9 g Al, 64 g O_2), slides 21–22
Homework match: HW23 — 5.65 g of each reactant: determine the limiting reactant (S + H_2SO_4; MnO_2 + H_2SO_4)
What it tests: Convert both reactants to moles, then use the mole ratio to see which one runs out first.

$$\text{Al: } \frac{17.9}{26.98} = 0.663 \text{ mol}; \qquad O_2\text{: } \frac{64}{32.00} = 2.0 \text{ mol}$$

O_2 needed for all the Al: $0.663 \times (3/4) = 0.498$ mol. That is less than the 2.0 mol available, so **Al runs out first and is the limiting reactant**.

Why the other choices are wrong:

- a) Al_2O_3 is the limiting reactant: A product can never be the limiting reactant.
- c) Neither; the reactants are present in exactly the stoichiometric ratio: 0.663 mol Al : 2.0 mol O_2 is not 4 : 3.
- d) O_2 is the limiting reactant: You need only 0.498 mol O_2 and there are 2.0 mol, so O_2 is in excess.
- e) O_2 is the limiting reactant because its coefficient is smaller: Coefficients alone don't decide it. Compare moles available with moles needed.

34. Theoretical yield with a limiting reactant — c) 160 g

Source: Chapter 9 post, slide 27 (Theoretical Yield Practice 2: 68 g Na + 97 g Cl_2), slides 25–26
Homework match: HW24 — hw24ex2.png (theoretical mass of XeF_4 from 173 g Xe + 104 g F_2)
What it tests: Find the limiting reactant, then calculate the product from it.

$$\text{Na: } \frac{68}{22.99} = 2.958 \text{ mol}; \qquad \text{Cl}_2\text{: } \frac{97}{70.90} = 1.368 \text{ mol}$$

Na needed for all the Cl_2: $1.368 \times 2 = 2.736$ mol. You have 2.958 mol, so Na is in excess and **Cl_2 is limiting**.

$$1.368 \text{ mol Cl}_2 \times \frac{2 \text{ mol NaCl}}{1 \text{ mol Cl}_2} \times \frac{58.44 \text{ g}}{\text{mol}} = 160 \text{ g NaCl}$$

Why the other choices are wrong:

- a) 2.74 g: That is the moles of NaCl, not grams.
- b) 80 g: Cl_2 limiting but used a 1:1 ratio (forgot the 2 NaCl).
- d) 165 g: Added the reactant masses. Na is left over, so it doesn't all become product.
- e) 173 g: Used Na as the limiting reactant (Cl_2 actually runs out first).

35. Percent yield — c) 77.4%

Source: Chapter 9 post, slide 32 (Percent Yield Practice 2: 191 g $KClO_3$, 89.9 g KCl), slide 30 (formula), slide 29
Homework match: HW24 — hw24ex1.png (percent yield of magnesium oxide; $Na_2S_2O_3 \cdot 5H_2O$)
What it tests: Theoretical yield by stoichiometry, then (actual / theoretical) × 100%.

$$191 \text{ g KClO}_3 \times \frac{1 \text{ mol}}{122.55 \text{ g}} \times \frac{2 \text{ mol KCl}}{2 \text{ mol KClO}_3} \times \frac{74.55 \text{ g}}{\text{mol}} = 116.2 \text{ g KCl (theoretical)}$$

$$\%\text{ yield} = \frac{89.9}{116.2} \times 100\% = 77.4\%$$

Why the other choices are wrong:

- a) 47.1%: Divided by the mass of reactant, not the theoretical yield of KCl.
- b) 51.6%: Used the 3 from O_2 in the ratio (3/2 instead of 2/2).
- d) 116%: That number is the theoretical yield in grams, not a percent.
- e) 129%: Upside down: theoretical / actual. Percent yield can't be over 100%.